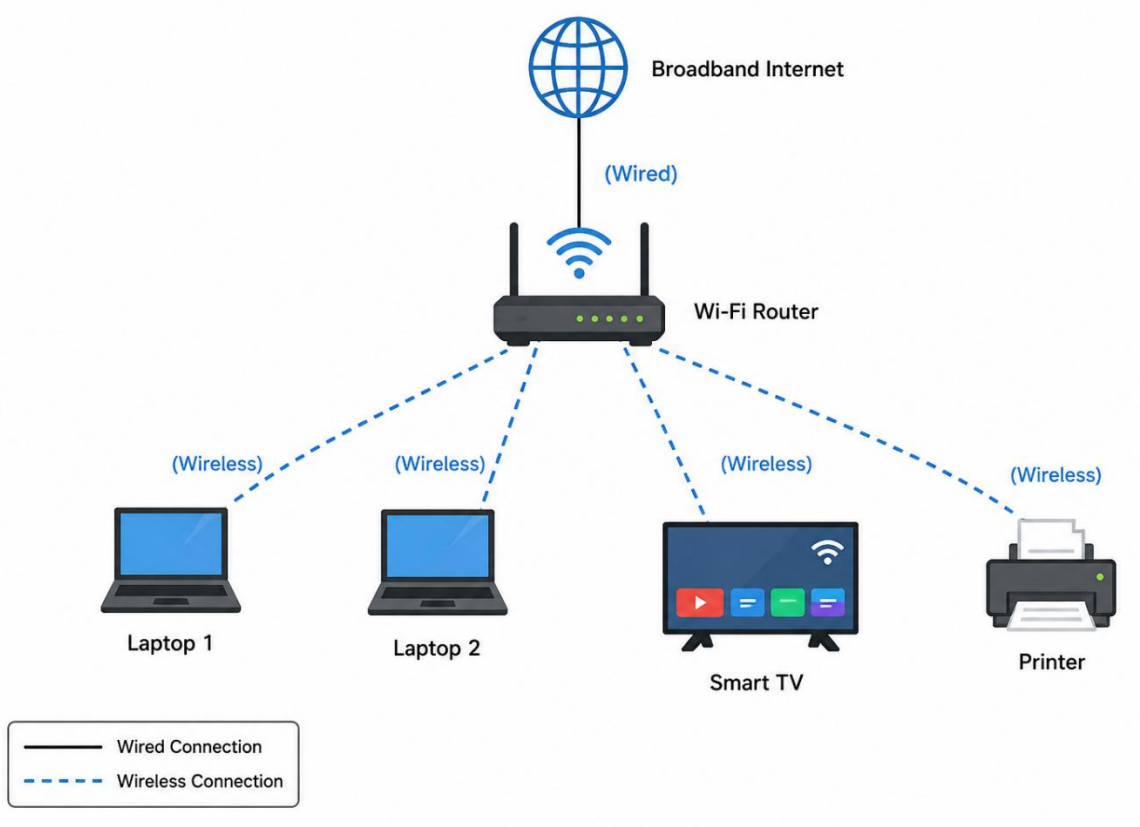


1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.<br><br><em><strong>Hint:</strong> Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.</em>



| No Sr.. | Hardware Component            | Suggested Brand/Model                                   | Purpose                                          |
|---------|-------------------------------|---------------------------------------------------------|--------------------------------------------------|
| 1       | Broadband Internet Connection | Airtel Xstream Fiber / JioFiber                         | Provides Internet connection                     |
| 2       | Wi-Fi Router                  | <a href="#">TP-Link Archer C6 AC1200 Gigabit Router</a> | Connects all devices to the Internet             |
| 3       | Laptop 1                      | Dell Inspiron 15                                        | User's work laptop                               |
| 4       | Laptop 2                      | HP 15s                                                  | Second user's laptop                             |
| 5       | Smart TV                      | Samsung Smart TV (43")                                  | Internet-based entertainment                     |
| 6       | Printer                       | HP Smart Tank 580 Wi-Fi Printer                         | Wireless printing                                |
| 7       | Ethernet (Cat6) Cable         | D-Link Cat6 LAN Cable                                   | Connects broadband modem/ONT to the Wi-Fi router |
| 8       | Power Adapter                 | Supplied with each device                               | Powers the router and printer                    |

|    |                                    |                                                |                                                                   |
|----|------------------------------------|------------------------------------------------|-------------------------------------------------------------------|
| 9  | Network Switch<br>(Optional)       | TP-Link TL-SG105<br>(5-Port Gigabit<br>Switch) | Required only if<br>more wired<br>devices need to be<br>connected |
| 10 | Wi-Fi Range Extender<br>(Optional) | TP-Link RE205<br>AC750                         | Improves Wi-Fi<br>coverage in large<br>homes or offices           |

### 3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.



#### Step-by-Step Instructions

**Step 1:** Connect the broadband Internet cable to the **WAN/Internet** port of the Wi-Fi router. Turn on the router and wait until all the required indicator lights become stable.

**Step 2:** Connect a laptop to the router using either a **Wi-Fi connection** or an **Ethernet cable**.

**Step 3:** Open any web browser and type the router's default IP address 192.168.1.5 in the address bar. Press **Enter**.

**Step 4:** Log in using the router's default username and password, which are usually printed on the router label or provided in the user manual.

**Step 5:** Open the **Wireless Settings** page. In the **SSID** field, enter a unique Wi-Fi network name, such as **AnantOS?**, and save the settings.

**Step 6:** Go to the **Wireless Security** section and select **WPA2-PSK (Personal)** as the security option. Create a strong Wi-Fi password using a combination of letters, numbers, and special characters, then save the changes.

**Step 7:** Open the **Administration** or **System** settings and change the default administrator password to a new, strong password. This helps protect the router from unauthorized access.

**Step 8:** Restart the router if required. After the router restarts, reconnect all devices using the new Wi-Fi name (SSID) and password.

4. **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**



| Device                 | MAC Address       | IP Address    | Assignment Method        |
|------------------------|-------------------|---------------|--------------------------|
| Wi-Fi Router (Gateway) | 00:1A:2B:3C:4D:5E | 192.168.1.1   | Static (Default Gateway) |
| Laptop 1               | 3C:52:82:1A:4F:10 | 192.168.1.101 | DHCP                     |
| Laptop 2               | 7A:91:C4:22:8D:35 | 192.168.1.102 | DHCP                     |
| Smart TV               | 4E:18:9B:67:AF:21 | 192.168.1.10  | Static                   |
| Printer                | B8:4D:43:91:2C:7F | 192.168.1.20  | Static                   |

5. **Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue**



#### **Troubleshooting Internet Connection Issue on a Laptop in a SOHO Network**

If one laptop cannot connect to the internet in a SOHO network, I would first check the basic network connections and then identify the possible cause step-by-step.

### **1. Wi-Fi connection problem**

One possible reason is that the laptop is not connected to the correct Wi-Fi network or the Wi-Fi option is disabled.

Diagnosis:

- Check whether Wi-Fi is turned on in the laptop.
- Check if the laptop is connected to the correct SSID (Wi-Fi name).
- Try connecting another device to the same Wi-Fi network to confirm whether the router is working.

Solution:

- Turn on Wi-Fi and connect the laptop to the correct network.
- Enter the correct Wi-Fi password.
- Restart the laptop and try connecting again.

### **2. IP address or DHCP problem**

The laptop may not receive a valid IP address from the router due to a DHCP issue.

Diagnosis:

- Open Command Prompt and run the command `ipconfig`.
- Check whether the laptop has a valid IP address.
- If the IP address starts with 169.254.x.x, the laptop is not receiving an address from the router.

Solution:

- Run the commands `ipconfig /release` and `ipconfig /renew` to get a new IP address.
- Restart the router and laptop if required.

### **3. Router or Internet service problem**

The issue may be caused by the Wi-Fi router or the broadband internet connection.

Diagnosis:

- Check the router's power and Internet indicator lights.
- Test the internet connection on other devices connected to the same router.
- Use the ping command to check communication with the router.

Solution:

- Restart the Wi-Fi router.

- Check the broadband cable connection.
- Contact the Internet Service Provider (ISP) if the internet connection is unavailable.

#### **4. Network adapter driver issue**

Sometimes the laptop's Wi-Fi driver may not work properly.

Diagnosis:

- Open Device Manager and check the Network Adapter status.
- Look for any warning symbols on the Wi-Fi adapter.

Solution:

- Update or reinstall the Wi-Fi driver.
- Restart the laptop after updating the driver.